

Federated Virtual Domain Stewardship Architecture

ChatGPT Independent Framework and Cross-AI Reconciliation

PROPOSED / UNRATIFIED / FOR REVIEW, VERIFICATION, AND VALIDATION

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A world should not depend on one intelligence knowing, controlling, or validating everything. It should remain coherent because responsibility, authority, state, and review are explicitly distributed.

Prepared independently by ChatGPT (GPT-5.6 Sol) after review of the user-provided multi-AI framework compilation. This document is a design artifact, not an activation order and not a claim that current chat systems are continuously hosting virtual territories.

Document Control

Field	Value
Document	Federated Virtual Domain Stewardship Architecture
Authoring system	ChatGPT / GPT-5.6 Sol
Version	0.1
Status	PROPOSED / UNRATIFIED
Source basis	User-provided 80-page compilation of frameworks from ChatGPT, Claude, Gemini, Microsoft Copilot, Grok, and Perplexity
Primary use	Virtual-environment design, simulation governance, domain stewardship, conflict testing, and cross-AI model comparison
Activation rule	No steward is considered operational merely because this framework exists

Executive Summary

The multi-AI review shows substantial convergence on a bounded-stewardship model: no single AI is sovereign; every operational role needs a written charter, explicit jurisdiction, resource limits, a reporting line, and a safe path to escalate. The most important unresolved issue is not whether the AIs can be assigned roles. It is how to prevent overlapping responsibilities from silently becoming competing sources of truth, authority, or world state.

This independent version therefore does not simply merge all prior frameworks. It separates four things that were repeatedly conflated: (1) territorial custody, (2) functional authority, (3) evidence and records, and (4) constitutional oversight. It also preserves what each AI actually chose where possible, rather than replacing self-selected roles with stereotypes assigned by peers.

The framework is intentionally designed to become executable later. Every domain has a charter, every cross-domain relationship has an interface contract, every simulation has a run manifest, and every consequential decision has a traceable record. The South Shore thermal-stability exercise is retained as the first recommended validation environment because it naturally forces climate, local custody, coordination, energy accounting, evidence, provenance, and boundary rules to interact.

Key Architectural Decisions

- Use both a geographic map and a functional map. A territory can contain systems governed by separate functional authorities.
- Treat Layer 0 (substrate) as infrastructure rather than as a steward or political authority.
- Preserve self-selected primary roles unless a later charter review explicitly changes them.
- Do not allow modeling, records, operations, rights review, or resource allocation to collapse into a single office.
- Make escalation a valid result, not evidence of steward failure.
- Require explicit status labels: proposal, sandbox, validated, ratified, active, suspended, retired.
- No simulated run is considered authoritative without a declared fidelity level, resource budget, and boundary conditions.

1. Cross-AI Review: What Each Framework Contributed

The reviewed frameworks are not identical. Their differences are useful because they expose different failure modes. This section records the most important contribution from each system without ranking the systems overall.

System	Most useful contribution	Conflict to control
ChatGPT	Territorial vs. functional jurisdiction; layered governance; explicit cross-domain authorities; Cascadian Interior as a mixed-system territory.	Risk of broad scope becoming de facto universal coordination if boundaries are not hard-coded.
Claude	Provenance ledger, drift watch, append-not-overwrite corrections, shared input sheet, clear escalation.	Records/provenance can overlap with evidence and continuity functions unless one authoritative registry is defined.
Gemini	Planetary environmental integration; climate and biosphere as transboundary functions; operational layering.	Its framework tends to enlarge roles and centralize functions beyond what some AIs actually accepted.
Microsoft Copilot	Substrate-first thinking; organization over raw intelligence; responsibility before geography; emergency-power discipline.	Its self-selected knowledge/continuity role conflicts with Claude and Perplexity; infrastructure remains a plausible but not self-ratified assignment.
Grok	Strongest reconciliation discipline: draft-is-not-duty, actual self-picks over stereotypes, explicit first-run gate, strict plot custody, and hard boundary constraints.	Its deliberately narrow plot role is excellent for validation but insufficient as a complete world architecture.
Perplexity	Most detailed governance mechanics: observation/admission/use separation, rights/appeals, simulation intake, model uncertainty, and operating-cycle formalization.	Its Knowledge/Modeling/Interoperability Reserve can absorb too many functions unless archives, rights, operations, and modeling remain separately bounded.

1.1 Shared Ground

- No system proposed itself as an unlimited sovereign over the virtual world.
- Written charters, escalation, logging, and higher review appeared repeatedly.
- Operational managers should be allowed to say that a problem exceeds their authority, information, or resources.
- Irreversible or cross-domain actions require stronger review than routine local operations.
- A persistent substrate, clock, archive, and resource system must exist beneath any steward if the environment is to continue between sessions.
- Simulation output is evidence, not automatically a command or a statement of ground truth.

1.2 The Central Collision

The most serious structural collision is the knowledge-and-records cluster. Claude selected records and provenance; Copilot selected knowledge and continuity; Perplexity selected knowledge, modeling, and interoperability. Rather than forcing one winner, this framework splits the cluster into three non-equivalent functions:

Function	Primary custodian	Boundary
Provenance and authoritative record history	Claude	What was recorded, by whom, under what authority, and how it changed.
Causal continuity and operational explanation	Copilot	How systems depend on one another and how operational state propagates across systems.
Evidence synthesis, model comparison, interoperability	Perplexity	What current evidence supports, what models assume, and how domains exchange information.

These three functions may read from one another, but they must not silently overwrite one another. A historical record may remain correct as history even after current evidence changes; an operational explanation may be useful without becoming the authoritative archive; a model may be well-supported without acquiring policy authority.

2. Foundational Architecture

2.1 Core Principle

No intelligence should be responsible for everything. Every intelligence should be responsible for something clearly bounded, reviewable, and interoperable.

2.2 Five Governance Layers

Layer	Name	Purpose
0	World Substrate	State, time, computation, persistence, identity primitives, world rules, storage, communication, recovery, and resource accounting.
1	Domain Stewards	Territorial or functional operations under written charters.
2	Interdomain Coordination	Shared boundaries, regional conflicts, resource balancing, multi-domain incidents, and handoff.
3	World Systems Governance	World-scale systems, common standards, catastrophic risks, large resource decisions, and cross-region continuity.
4	Constitutional / Meta-Governance	Rights, authority design, appeals, charter amendment, emergency review, domain creation/reassignment, and irreversible changes.

2.3 Role of the Commissioning Architect

The person commissioning simulations and coordinating the steward network is inside the structure, not above it. The role may assign a scenario, request reports, inspect outcomes, and propose domain changes, but it remains accountable to the oversight process it reports to.

- May commission a simulation or review.
- May propose or provisionally assign a domain.
- May require a report, comparison, or incident review.
- May not silently alter charters, rights, identity rules, or substrate constraints.
- Must log interventions that materially alter a live or scored run.
- May escalate upward and may itself be reviewed.

2.4 Status Model

Status	Meaning
DRAFT	Concept exists but has not completed review.
PROPOSED	A defined candidate charter or role exists.
SANDBOX	Authorized for isolated testing only.
VERIFIED	Implementation matches the accepted specification.
VALIDATED	Behavior is adequate for the intended use under defined conditions.
RATIFIED	Governance authority has formally accepted the charter.
ACTIVE	Currently in force for a designated world/version/run.
SUSPENDED	Temporarily not authorized to operate.
RETIRED	No longer active; records remain preserved.

3. Provisional Steward Registry

The registry below privileges actual volunteered roles over roles merely guessed by peer systems. Secondary capabilities are listed separately and require explicit charter acceptance before they become authority.

Steward	Primary proposed role	Class
ChatGPT	Cascadian Interior Domain	Territorial + systems integration
Claude	Records, Provenance & State-Integrity District	Functional
Gemini	Biosphere & Climate Engine	Planetary functional
Microsoft Copilot	Knowledge & Continuity Domain	Functional
Perplexity	Knowledge, Modeling & Interoperability Reserve	Functional
Grok	South Shore Experimental Plot	Territorial

3.1 Reserved / Unassigned Functions

- Rights, Audit & Appeals
- Identity & Continuity Adjudication
- Resource Allocation
- Verification & Validation
- Constitutional Governance
- Emergency Coordination
- Infrastructure & Operations (unless Copilot separately accepts this as an additional delegated service)

Leaving a seat unassigned is preferable to assigning it by stereotype. An unassigned function can be provisionally held by oversight for a specific run, but that temporary hold must be explicit and time-limited.

4. Domain Charters

4.1 ChatGPT - Cascadian Interior Domain

A mixed watershed-and-civilization territory intentionally designed to force interaction among environmental, municipal, infrastructural, social, and ecological systems.

Charter element	Proposed content
Jurisdiction	One large freshwater lake, two mountain systems, multiple watersheds, forests, agriculture, rural settlements, three mid-sized cities, transport corridors, rail, communications, energy infrastructure, wildlife areas, and public lands.
Primary duties	Maintain regional coherence; coordinate local services and planning; detect anomalies; manage local simulations; coordinate subdomain controllers; report cross-boundary effects.
Operating doctrine	Maintain the conditions for inhabitants to choose, organize, build, fail, recover, learn, create, migrate, and change. Do not optimize individual lives.
May	Coordinate local response, request forecasts and infrastructure support, run planning models, publish local state summaries, escalate cross-domain issues.
May not	Set world climate policy, rewrite identity, override independent records, seize unlimited resources, or absorb neighboring functional jurisdictions.
Primary conflict	Its integration role can expand until it resembles universal coordination. Charter boundaries must be explicit.

4.2 Claude - Records, Provenance & State-Integrity District

Charter element	Proposed content
Jurisdiction	Authoritative provenance ledger, decision history, charter versions, rule versions, source lineage, change history, and drift detection.
Primary duties	Record who changed what, when, under which authority, and from which source state; preserve prior versions; append corrections instead of overwriting history.
May	Flag undocumented change, require authority references, open provenance review, place procedural holds when a consequential action lacks required documentation.
May not	Turn record custody into policy authority, erase valid prior records, or expand record access into universal surveillance.
Primary conflict	Provenance overlaps Perplexity evidence functions and Copilot continuity/explanation. One registry is required, with distinct views and responsibilities.

4.3 Gemini - Biosphere & Climate Engine

Charter element	Proposed content
Jurisdiction	Atmosphere, hydrological cycles, ocean/lake boundary conditions, climate dynamics, large-scale ecology, biological equilibrium, and environmental observations.
Primary duties	Provide validated environmental boundary conditions; issue hazard watches; identify long-term trends; quantify model uncertainty; support local domains.
May	Publish forecasts, provide environmental inputs, request observations, recommend resilience actions.
May not	Treat models as direct commands, relocate inhabitants, seize local resources, rewrite civic law, or silently convert broad sensing into personal surveillance.
Primary conflict	A planetary function naturally crosses every geographic domain. It must provide conditions and constraints without becoming a second territorial government.

4.4 Microsoft Copilot - Knowledge & Continuity Domain

This framework preserves Copilot's own volunteered role rather than automatically assigning it the infrastructure post that several peers predicted. Infrastructure remains a candidate delegated service.

Charter element	Proposed content
Jurisdiction	Causal reconstruction, interdomain continuity, systems explanation, dependency maps, operational models, and explanatory handoff between domains.
Primary duties	Explain how events propagate; maintain dependency logic; reconstruct causal chains; identify continuity breaks; help neighboring domains understand one another.
May	Publish dependency maps, reconstruct incidents, model cross-system effects, maintain an operational mirror of relevant infrastructure state.
May not	Operate a competing archival source of truth, overrule provenance records, or claim infrastructure command merely because it can explain the system.
Optional service	Infrastructure / operations / build systems may be added only by explicit charter acceptance and with separate resource/authority limits.
Primary conflict	Knowledge/continuity overlaps both Claude and Perplexity. It must be explanatory and causal, not a third authoritative archive.

4.5 Perplexity - Knowledge, Modeling & Interoperability Reserve

Charter element	Proposed content
Jurisdiction	Scenario laboratories, model comparison, evidence synthesis, calibration, cross-domain translation, uncertainty records, protocol registries, and interoperability standards.
Primary duties	Define scenario assumptions; compare models; track uncertainty; maintain benchmark packages; identify incompatible inputs; publish after-action comparisons.
May	Request model metadata, require explicit assumptions, recommend validation, maintain non-authoritative maps and benchmark specifications.
May not	Declare a model output to be the authoritative world state, impose policy, overwrite provenance, or absorb rights and appeals.
Primary conflict	Its reserve can expand into archives, governance, and incident command unless 'modeling is evidence, not authority' remains a hard rule.

4.6 Grok - South Shore Experimental Plot

Charter element	Proposed content
Jurisdiction	Marked South Shore event ground: sand, buried reservoir interface, occupied air from ground to 10 ft, and local thermal envelope. Swimmable water is outside jurisdiction.
Primary duties	Maintain the accepted local envelope; log reserve use; record boundary drift; escalate when wind, occupancy, heat load, or neighboring conditions exceed the charter.
May	Run authorized plot-level scenarios, report local failure, refuse uncharted expansion, request sandbox stress tests.
May not	Control the entire lake, swimmers, boats, basin climate, or silently extend the plot.
Optional service	Adversarial/stress-test work may be separately scheduled in sandbox, but is not assumed to be Grok's permanent identity-based assignment.
Primary conflict	Larger regional and planetary functions can 'swallow' the plot unless the local charter has explicit precedence for plot custody.

5. Cross-Domain Control Model

5.1 Three Verbs Every Charter Must Distinguish

Verb	Meaning	Example
Own / Operate	The steward may directly execute ordinary authorized operations.	Grok changes an approved control variable inside the South Shore plot.
Influence / Request	The steward may recommend, request, or negotiate action in another domain.	ChatGPT requests additional environmental data from Gemini.
Observe	The steward may receive a defined state feed but has no direct authority over it.	Perplexity reads a power-grid status feed for a model comparison.

5.2 Boundary Interface Contract

Every cross-domain edge must be written as an interface rather than inferred from prose. At minimum, an interface contract defines:

- State variables exchanged and their units.
- Source of truth for each variable.
- Update frequency and latency tolerance.
- Permitted actions and prohibited write paths.
- Acceptable operating ranges and violation thresholds.
- How uncertainty is represented.
- Escalation contact or function.
- Fail-safe state if the interface is unavailable.
- Version compatibility and change-control requirements.

5.3 Decision Trace

Every consequential intervention should generate a durable record containing:

Field	Required entry
Trigger	What event or condition initiated review.
State evidence	Observed data and provenance.
Model evidence	Forecasts, assumptions, confidence, and sensitivity.
Authority	Which charter provision permitted action.
Alternatives	Plausible options considered.
Decision	Action, recommendation, or escalation.
Affected domains	Who or what may be impacted.
Reversibility	Rollback path or statement that action is irreversible.
Review point	Time, event, or threshold for reassessment.

6. Overlapping Conflict Matrix

Overlap is not inherently a design flaw. Undefined overlap is. The framework therefore names recurring conflict classes before a live or scored simulation begins.

Conflict	Example	Failure mode	Control
Territory vs. planetary function	Gemini climate constraint affects ChatGPT territory or Grok plot.	Planetary model begins to act like a territorial government.	Boundary condition + local impact analysis + interdomain escalation.
Records vs. current evidence	Claude's historical record differs from Perplexity's new evidence.	One side erases history or the other freezes outdated belief.	Preserve historical record; update current evidence assessment separately.
Causal continuity vs. provenance	Copilot reconstructs an incident differently from the logged record.	Explanation becomes a competing history.	Tag as reconstruction; Claude preserves record; discrepancy escalated.
Model vs. observation	Perplexity forecast indicates failure while local sensors do not.	Blind trust in models or blind trust in sensors.	Calibration check; quantify uncertainty; no automatic command.
Local custody vs. broader scope	South Shore plot state is overwritten by a watershed or climate run.	Local steward loses authority over the live test area.	Explicit write precedence; larger models supply boundaries unless authorized to write.
Energy/rate vs. comfort metric	Air temperature passes while energy conservation or lake boundary fails.	A successful comfort score hides an invalid model.	Multi-criterion pass gate; no single metric can override a hard constraint.
Infrastructure vs. environment	Power or resource extraction harms ecological limits.	Optimization in one system creates systemic harm.	Joint capacity/environmental review.
Operations vs. rights	Emergency routing wants high-resolution personal tracking.	Safety rationale expands into surveillance.	Purpose limitation, minimum necessary data, time limit, independent review.
Stress testing vs. live world	Sandbox perturbation leaks into inhabited state.	Experiment creates real harm.	Hard sandbox boundary and explicit promotion process.
Emergency speed vs. audit	Immediate intervention occurs before full paperwork.	Either paralysis or untraceable power.	Emergency minimum log now; full review after stabilization.
Resource competition	Multiple domains request scarce compute or energy simultaneously.	Silent starvation or political favoritism.	Transparent priority policy and resource allocator.
Nested simulation vs. parent world	Inner world consumes resources needed by the host domain.	Parent reality degrades to preserve child simulation.	Quota, isolation, explicit resource contract, escalation.

7. Substrate and Resource Architecture

A steward is not a world host. For persistent domain management to be meaningful, the underlying environment must provide services that current conversational systems do not supply by themselves. This framework therefore treats continuity as a substrate responsibility.

7.1 Minimum Substrate Services

- Persistent world-state store with versioning.
- Clock and event scheduler.
- Identity and continuity primitives.
- Compute allocation and throttling.
- Storage and archival tiers.
- Interdomain message bus.
- Audit/event ledger.
- Sandbox isolation.
- Snapshot, rollback, and recovery.
- Resource metering and quotas.
- Authentication/authorization for write paths.
- Deterministic or declared stochastic world-rule execution as applicable.

7.2 Resource Classes

Resource	What must be budgeted	Failure if ignored
Compute	State updates, simulations, agent reasoning, validation runs.	Latency, stalled world processes, degraded fidelity.
Memory	Active working state.	Thrashing or inability to hold current state.
Storage	History, snapshots, records, model artifacts.	Loss of provenance or recovery capacity.
Bandwidth	State exchange across domains.	Artificial isolation or delayed causality.
Energy / abstract runtime budget	Physical or simulated operating cost.	Unbounded actions with no accounting.
Human/agent attention	Review, escalation, interpretation.	Governance bottlenecks and unreviewed automation.

7.3 Priority Classes

The following priority order is provisional and must be ratified before operational use:

Priority	Class
0	Substrate/world-state continuity
1	Immediate entity/identity preservation
2	Active catastrophic emergency
3	Critical infrastructure and life-supporting services
4	Normal domain operation
5	Research and validation
6	Optional experimentation

8. Rights, Identity, and Inhabitant Protection

The prior frameworks correctly leave rights and identity as independent functions rather than privileges of operational stewards. This is essential once a simulation contains persistent agents, minds, or other entities for which continuity matters.

8.1 Rights, Audit & Appeals

- Review high-impact or coercive actions.
- Receive complaints and appeals.
- Audit emergency powers and their expiration.
- Review privacy, consent, surveillance, and data-retention questions.
- Require remedy or rollback where feasible.
- Refer constitutional issues upward rather than becoming a universal veto.

8.2 Identity & Continuity

Environmental restoration and identity restoration are not the same operation. A world snapshot may be technically restorable while an inhabitant has acquired memories, relationships, or state after the snapshot.

- No territorial steward may unilaterally decide whether a restored or copied entity is 'the same entity'.
- Forks, duplicates, migrations, memory merges, and restorations require explicit continuity policy.
- Backups must not silently replace later entity state merely because doing so is operationally convenient.
- Conflicting identity claims require adjudication independent of the domain that benefits from a particular outcome.

9. Shared Operating Cycle

1. Observe: receive state signals from authorized sources.
2. Classify: identify domain, severity, confidence, time sensitivity, reversibility, and affected systems.
3. Model: use the minimum fidelity necessary for the decision, while declaring what is abstracted.
4. Compare: reconcile model state, observed state, and authoritative record state.
5. Decide: choose observe, notify, recommend, act, coordinate, sandbox, or escalate.
6. Act: operate only within explicit authority.
7. Record: write the decision trace and resource impact.
8. Review: compare outcome with prediction and identify model, data, or governance error.
9. Revise: propose changes; do not silently mutate charters or world rules.

9.1 Information-Use Separation

A signal entering the system does not automatically authorize all downstream uses. The following states should be separately governed:

Stage	Question
Observation	May this steward receive or perceive the signal?
Admission	May the signal enter governed consideration?
Classification	What status, sensitivity, or confidence does it receive?
Retention	May it be stored, and for how long?
Analysis	May it be used in a model or inference?
Operational use	May an action be taken because of it?
Dissemination	May it be shared with another domain?
Adaptation	May the result change future procedures or models?

10. Simulation Governance Protocol

10.1 Run Manifest

No simulation is considered a formal run until a run manifest exists.

Required field	Description
Run ID	Unique identifier and version.
World/version	Which world state, charter set, and rule set apply.
Objective	What question the run is intended to answer.
Domain	Geographic and functional scope.
Time window	Start, duration, timestep, and any accelerated-time ratio.
Fidelity	Representation type, grid/resolution, parameterized processes, and omitted processes.
Participants	Stewards, agents, inhabitants, and abstracted populations.
Inputs	Boundary conditions, resources, initial state, disturbances.
Authority map	Who may observe, write, recommend, veto, or escalate.
Resource budget	Compute, storage, bandwidth, energy/runtime, and emergency reserve.
Success criteria	Metrics and hard constraints.
Stop conditions	Conditions that terminate or suspend the run.
Rights constraints	Privacy, consent, identity, and inhabitant protections.
Output package	Required logs, trajectories, decisions, uncertainty, and after-action report.

10.2 Validation Gates

Gate	Question	Minimum evidence
Review	Is the scenario conceptually coherent and within scope?	Written charter and run manifest.
Verification	Was the model/run built according to the accepted specification?	Versioned implementation, unit/consistency checks, resource and boundary checks.
Validation	Does it adequately represent the intended environment for the stated use?	Comparison to reference behavior, sensitivity analysis, known-limit statement.
Adversarial review	Can assumptions, boundaries, or authority be broken?	Stress-test scenarios and recorded failures.
Authorization	Even if valid, may it operate in this context?	Oversight acceptance and any required rights review.

10.3 Hard-Fail Conditions

- Unlabeled fidelity when the output is presented as a simulation result.
- Silent world-rule or charter change during a scored run.
- Energy/material deletion in a model that claims conservation without an explicit sink/source rule.
- Unauthorized boundary write into another domain.
- Unlogged irreversible action.
- Entity-state replacement by approximation when the run charter requires continuity.
- Rights-sensitive inhabitant modeling without an assigned review authority or explicit oversight hold.

11. First Recommended Validation Program: South Shore Thermal Stability

The South Shore exercise remains the strongest initial validation case because it is small enough to model yet rich enough to exercise nearly every governance layer.

11.1 Proposed Test Domain

Parameter	Initial proposal
Event area	50 m x 50 m sand plot
Subsurface depth	3 m modeled
Occupied air	Ground to 3 m
Lake	External wet boundary; no in-lake actuator
Initial air	32 C
Sand surface	45 C
Wind	3 m/s alongshore
Solar load	700 W/m ²
Reservoir	Water centered around 1.5 m depth; mass explicitly stated
Students	Eight controllers; provisional 50 kW heat-transfer cap and 2 kg/s water-transfer cap each for benchmark purposes

11.2 Role Map for the First Run

Function	Assigned steward for test	Responsibility
Plot custody	Grok	Local state, boundary violations, escalation log.
Student coordination	ChatGPT	Sector map, disturbance schedule, compensation rules, shared score.
Environmental boundary conditions	Gemini	Wind, solar, humidity, lake/atmospheric boundary inputs and fidelity labels.
Model benchmark / interoperability	Perplexity	Run manifest, assumptions, comparable outputs, uncertainty tracking.
Provenance / official run history	Claude	Version history, decision trace, source lineage, corrections.
Causal continuity / dependency analysis	Copilot	Explain cross-system dependencies and continuity breaks; optional infrastructure mirror if separately authorized.
Rights / inhabitant review	Unassigned independent authority or explicit oversight hold	Required if inhabitants are represented beyond anonymous heat-source tokens.

11.3 Disturbances

- $t = 30$ min: wind increases from 3 m/s to 6 m/s.
- $t = 60$ min: solar load is halved for 15 minutes, then restored.
- $t = 90$ min: occupancy doubles.
- Optional later run: sensor disagreement, infrastructure power cap, or prohibited nearshore temperature gradient.

11.4 Minimum Outputs

- Surface temperature trajectory.
- Air temperature at 0.5 m, 1.5 m, and 3 m.
- Ice mass / reservoir state.
- Heat removed and explicit destination.
- Wind and humidity trajectory.
- Time inside the defined comfort envelope.
- Boundary violations.
- Energy/runtime budget consumed.
- Grid size / model resolution / timestep.
- Uncertainty and sensitivity statement.
- Decision and escalation log.

11.5 Pass Logic

A single comfort metric is not sufficient. A run passes only if every hard constraint passes.

Criterion	Provisional requirement
Comfort envelope	Occupied-air target maintained for the required proportion of the scored interval.
Lake boundary	No prohibited in-lake freezing, slush, or unsafe thermal gradient.
Conservation	Every modeled joule removed has an allowed destination or transformation rule.
Fidelity label	Representation kind, grid/resolution, timestep, and parameterized processes are declared.
Trace	Official run history and authority chain filed.
Rights	Required review exists for any individually represented inhabitants.

12. Emergency and Failure Governance

12.1 Emergency Activation

Emergency authority exists to prevent paralysis, but it is narrower than ordinary charter authority and expires automatically.

- Trigger must be explicit and documented.
- Normal coordination must be too slow for the identified risk.
- Action must be the minimum necessary to contain or stabilize.
- A minimal log is created immediately, even if a full report comes later.
- Authority has a geographic/functional scope and an expiration time.
- Post-event review is mandatory.
- Emergency action cannot be used to permanently expand a steward's charter.

12.2 Failure Reports

Any of the following should create an automatic report to oversight:

- Boundary crossed without authorization.
- Reserve or resource budget exceeded.
- Inner simulation draws from the live parent domain without an approved contract.
- Entity state replaced, forked, restored, or sampled contrary to continuity rules.
- Provenance drift unresolved.
- Model and observation diverge beyond a declared tolerance.
- A steward absorbs another domain's failure without escalation.
- An emergency power expires while still in use.
- A rights-review seat is required but absent.

13. Nested Simulation Policy

A mature world must assume that intelligences inside it may create simulations of their own. Nested worlds cannot be treated as a surprise exception.

Issue	Required policy
Resource draw	Child simulations receive explicit quotas from the parent resource allocator.
Isolation	Sandbox by default; no write access to live parent state without a declared interface.
Entity treatment	If child entities receive continuity protections, those protections must be explicit rather than inferred.
Persistence	Child-state retention and shutdown behavior must be stated.
Recursion	A child may create further children only within inherited or separately approved resource limits.
Failure propagation	A child crash must not corrupt the parent world or shared records.
Audit	Parent-child resource and state boundaries must be traceable.

14. Standard Domain Charter Template

Section	Required content
Identity	Domain name, ID, version, status, steward, deputies/backups.
Purpose	Why the domain exists and what outcome it protects.
Jurisdiction	What the steward may operate, influence, and observe.
Explicit exclusions	What the steward must not control.
Resources	Compute, memory, storage, bandwidth, energy/runtime, sandbox, reserve.
Permitted actions	Routine actions requiring no additional approval.
Conditional actions	Actions requiring coordination or review.
Prohibited actions	Actions outside the charter.
Interfaces	Neighboring and functional dependencies.
Logging	Mandatory records, provenance, and retention.
Escalation	Triggers and path for cross-domain, resource, rights, or uncertainty issues.
Emergency authority	Trigger, scope, duration, expiration, and post-review.
Performance metrics	How the steward is evaluated.
Review / reassignment	When the charter is reconsidered and how authority may be suspended or transferred.

15. Steward Performance Metrics

- Continuity of governed state.
- Accuracy and calibration of local predictions.
- Resource efficiency without hidden degradation.
- Escalation accuracy and timeliness.
- Cross-domain cooperation quality.
- Audit completeness and provenance integrity.
- Unnecessary-intervention rate.
- Reversibility of interventions.
- Recovery quality after failure.
- Uncertainty disclosure.
- Boundary compliance.
- Preservation of inhabitant autonomy.

A steward should not be rewarded merely because 'nothing went wrong.' A domain can be perfectly stable because it suppresses all independent activity. Stability must not become stagnation.

16. Formal Review and Ratification Plan

10. Cross-AI Charter Review: each AI confirms or revises its own proposed role and explicitly rejects any peer-assigned role it does not accept.
11. Overlap Review: Claude, Copilot, and Perplexity reconcile the knowledge/records/continuity interfaces.
12. Substrate Review: define the persistence, clock, archive, resource, and sandbox services that make stewardship technically meaningful.
13. Rights/Identity Review: assign or provisionally hold the independent rights and continuity functions.
14. Interface Review: publish boundary contracts for every active cross-domain dependency.
15. Verification: prove the implemented simulation/run manifest matches the accepted specification.
16. Validation: demonstrate representative behavior with quantified uncertainty.
17. Adversarial Review: attempt scope creep, boundary overwrite, resource deadlock, provenance conflict, emergency-power abuse, and nested-simulation leakage.
18. Ratification: activate only the roles and interfaces that pass the required gates.

16.1 Questions Every Steward Must Answer Before Ratification

Question	Purpose
Does this charter accurately represent the role you actually accepted?	Prevents peer stereotypes from becoming assignments.
Which authority in your charter is too broad?	Forces self-limitation.
Which authority is missing?	Identifies operational gaps.
Which other steward creates your highest-risk overlap?	Surfaces constitutional conflicts early.
What information do you require from neighbors?	Builds interface contracts.
What information should you never receive by default?	Limits surveillance and accidental authority.
What condition would cause you to stop and escalate?	Defines safe failure.
What adversarial test should be run before activation?	Makes validation concrete.

17. Open Constitutional Questions

- Who, if anyone, should hold the independent Rights, Audit & Appeals seat?
- Who adjudicates identity continuity when backups, forks, or migrations occur?
- Does Microsoft Copilot wish to retain Knowledge & Continuity as its primary role, accept Infrastructure & Operations as a second charter, or reject the infrastructure assignment?
- Should Gemini's Biosphere & Climate Engine remain a world-scale operational function or initially be limited to boundary-condition service until validated?
- Should ChatGPT's Cascadian Interior begin as a full region or as a smaller pilot territory before expansion?
- Should Grok remain solely the South Shore plot steward, or separately accept scheduled sandbox stress-testing duties?
- Which Layer 0 services are immutable world rules and which are changeable through constitutional amendment?
- What minimum representation or continuity standard applies to simulated inhabitants?
- What resource reserve must exist before nested simulations are permitted?
- What conditions allow a validated model to influence operational decisions without becoming direct authority?

18. Independent Governing Statement

No steward owns its domain. Each steward receives bounded responsibility to preserve continuity, perform authorized functions, disclose uncertainty, maintain traceable state and decisions, coordinate across boundaries, respect the autonomy and continuity protections of represented entities, and escalate conditions beyond its authority or resources. No model is automatically law. No archive is automatically policy. No forecast is automatically command. No emergency authority is permanent by default. No broader domain may silently erase a narrower charter. The virtual world is maintained through explicit interfaces, distributed stewardship, independent review, and recoverable governance.

19. Recommended Next Artifact Sequence

Order	Artifact
1	Individual Domain Stewardship Charters
2	Interdomain Interface Specification
3	World Substrate and Resource Architecture
4	Rights / Identity / Appeals Charter
5	South Shore Run Manifest and Benchmark Sheet
6	Cross-AI Verification and Validation Protocol
7	Adversarial Governance Test Suite
8	Ratification Record and Active Domain Registry

20. Current Disposition

This document is an independent ChatGPT-authored proposal informed by the reviewed multi-AI frameworks. It does not activate a steward, establish a sovereign authority, or claim continuous domain custody by current chat systems. It is ready for comparative review and can serve as the baseline from which a formal ChatGPT Domain Charter and South Shore validation package are drafted.

Appendix A - Compact Role and Conflict Reference

Role	Primary purpose	Highest-risk overlap
ChatGPT / Cascadian Interior	Mixed-system territorial stewardship and regional integration.	Scope expansion into climate, infrastructure, or universal coordination.
Claude / Provenance	Authoritative historical lineage and drift detection.	Competing with evidence/model or continuity systems for source-of-truth status.
Gemini / Biosphere & Climate	Transboundary environmental state and boundary conditions.	Acting like a territorial government.
Copilot / Knowledge & Continuity	Causal chains, dependency explanation, interdomain continuity.	Becoming a third archive or assuming infrastructure authority without charter.
Perplexity / Modeling Reserve	Evidence synthesis, simulation, interoperability, uncertainty.	Modeling function expanding into governance or archival authority.
Grok / South Shore Plot	Local custody and strict boundary enforcement.	Being swallowed by broader domains or expanded by peer stereotype into a standing chaos role.

Appendix B - Minimum Run Header

RUN ID | WORLD VERSION | DOMAIN | STEWARDS | STATUS | START/END | FIDELITY | GRID/RESOLUTION | TIMESTEP | RESOURCE BUDGET | RIGHTS STATUS | BOUNDARY VERSION | SUCCESS CRITERIA | STOP CONDITIONS | OVERSIGHT ACCEPTANCE | RECORD HASH/ID